PRODUCT USER MANUAL

For Global Ocean OSTIA Near Real Time Level 4 SST Product

SST_GLO_SST-L4_NRT_OBSERVATIONS_010_001

Issue: 8.0

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MERCATOR OCEAN
INTERNATIONAL

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
DRAFT	2011/05/31	All	First draft version, based on L3EUR PUM draft by Francoise Orain. Provided for the preparation of the POR for MyOcean V1	Matthew Martin	
1.0	2011/10/25	All	New version for V2 updates	Matthew Martin	
2.0	2011/11/09		Use of latest template	L. Crosnier	
3.0	2013/02/13	All	Dissemination changes for MyOcean V3, and update to latest template	A Sellar, N McConnell	
3.1	2013/03/12		Updating to include the addition of lake ice into OSTIA NRT products	E. Blockley	
3.2	2013/11/14		Updating to include the addition of monthly and seasonal means of the OSTIA NRT products. Update	A. McLaren	L. Crosnier
3.3	2014/02/10	I.4 III.2.3	Added information on correction made to analysis error computation on 5 Feb 2014. Also corrected relaxation timescale information in III.2.3.	A. McLaren	L. Crosnier
3.4	2015/05/01	all	Change format to fit CMEMS graphical rules		L. Crosnier
3.5	2015/10/22	All	Changed MyOcean references to CMEMS.	S. Good	
3.6	2016/02/23		Update for new data sources	E. Fiedler	B. Hackett
3.7	2016/08/15	I.3, I.4, III.1, III.2.2	Reprocessing of period in early 2016 (revert to MetOp-A data)	E. Fieldler, B. Hackett	B. Hackett
3.8	2016/10/24	I.4	Change to reference data used for bias correction (VIIRS)	S. Good	B. Hackett
3.9	2017/09/18		Change data assimilation scheme used to make the product to NEMOVAR	S. Good	
4.0	2018/05/25	I.3. I.4, III.1	Remove old information about TAC structure, record change to Baltic and lake ice and update satellite data usage information.	S. Good	

Issue	Date	§	Description of Change	Author	Validated By
4.1	2019/01/17	I.4, III.1, IV.1, IV.4, VI.1-3	Documented upgraded feature resolution and use of SLSTR-A. Updated information about bias files.	S. Good	
4.2	2019/01/17	I.4	New entry in history of the product.	S. Good	C. Derval
4.3	2019/03/22	all	Review of the document	S. Good	
4.4	2019/11/06	I.4, III.1	Upstream data change	S. Good	
5.0	2022/06/16	All	New template ; upstream data change	S. Good	
6.0	2022/12/15	I.1, I.2, II.2, II.5.1	Handover of activities due to Brexit	D. Briand	A. Pisano
7.0	2025/01/23	All	New template; reverse handover of activities due to Brexit; revised temporal coverage and datasets	S. Good	Copernicus Marine Product management
8.0	2025/05/28	I.b	Ingestion of VIIRS on NOAA21 observations	S. Good, C. Mao	Copernicus Marine Product management

TABLE OF CONTENTS

RECORD TABLE 2
GLOSSARY AND ABBREVIATIONS 5
DATA ACCESS 6
INTRODUCTION 7
a) Summary 7
b) History of changes 7
DESCRIPTION OF THE PRODUCT SPECIFICATION 11
a) General Information 11
b) Production System Description 12
c) Dataset details 13
d) Additional information on some parameters 14
e) Processing information 14
f) Catalogue updating 14
g) Time averaging 14
h) Product use and impact 14
FILE FORMAT 15
FILES NOMENCLATURE 15
REFERENCES 16

GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
SST	Sea surface temperature
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

INTRODUCTION

a) Summary

This is the Product User Manual describing the SST-GLO-SST-L4-NRT-OBSERVATIONS-010-001 Copernicus Marine Service product: how it is built, the content, and how to access it.

The main objective of the Copernicus Marine Service project is to deliver and operate a rigorous, robust and sustainable Ocean Monitoring and Forecasting system to users for all marine applications: maritime safety, marine resources, marine and coastal environment and climate, seasonal and weather forecasting.

The Copernicus Marine Service system is composed of various sub-systems, among which are the Monitoring and Forecast Centers (MFCs), in charge of producing analyses and forecasts of the ocean state, and the Thematic Assembly Centers (TACs), in charge of producing satellite and in-situ observations-based products.

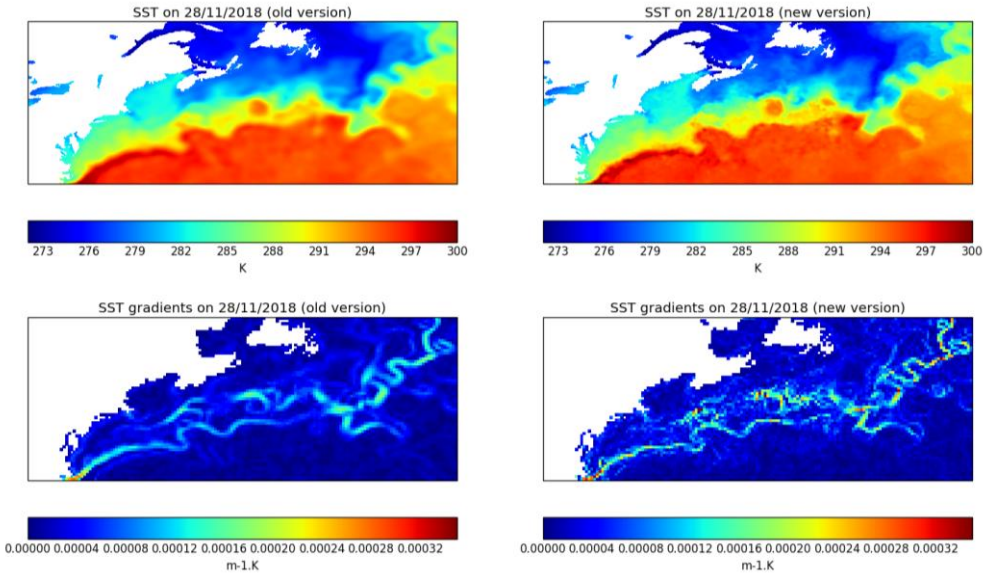
Copernicus Marine Service sea surface temperature (SST) satellite observations are managed by the Sea Surface Temperature (SST) TAC. Some quality monitoring and validation statistics of the SST TAC near real-time products are displayed at http://marine.copernicus.eu/services-portfolio/scientific-quality/.

b) History of changes

Date	Description of changes and impacted product
2012/01	Changed NetCDF format to the latest GHRSST standard (described in section III).

2013/01/17	A number of improvements were made, aimed at improving the quality of the product. The main impact was that OSTIA is now able to resolve finer scales. The changes were:
	• Update of the background error covariances in the OSTIA system:
	  o New updated seasonal background error variances are used for both the mesoscale and synoptic scale components of the background errors.
	  o Latitudinally varying anisotropic correlation length scales are used.
	  o For in situ observations, type dependant measurement errors have been implemented which are added to spatially varying representivity errors.
	  o The number of iterations of the assimilation has been increased from 10 to 100 iterations to ensure convergence.
	• Change in the minimum quality flag of the geostationary satellite SST data used in the OSTIA analysis (SEVIRI/MSG data; GOES-E data).
	• Set a minimum SST on the OSTIA SST analysis.
	• The reference data set used for the bias correction of the satellite data was expanded to include a subset of the most accurate MetOp AVHRR SST data (based on low satellite zenith angle). This replaces AATSR data which was included in the reference data set prior to its loss on 8th April 2012.

2013/04/17	Lake ice was added to OSTIA ice field using 2013 NCEP data for ice in inland waters. This change includes a slight relaxation to 0°C under lake ice but otherwise there is no effect on global ocean SST.
2013/11/11	A change was made to the OSTIA system to routinely produce monthly and seasonal mean files of the L4 product. In addition, the archive of NRT OSTIA data from 1st January 2007 to 15th February 2012 was changed to the current GDS2.0 file format, so that the data archive from 1st January 2007 onwards became directly available to users from the MyOcean SubSetter or FTP service.
2014/02/05	A change was made to OSTIA to correct the computation of the analysis error estimates in the system. This fixes a problem in the system that was present during the period 17 Jan 2013 to 4 Feb 2014 and will have caused the analysis error to be overestimated during this time. The SST analysis and sea ice fields were unaffected by this correction.
2016/02/23	The high quality subset of MetOp-A AVHRR observations used in the bias correction reference dataset was updated to MetOp-B AVHRR due to an upstream change in data availability.
2016/03/15	Two new global SST datasets were included in the OSTIA analysis: an infra-red dataset from the VIIRS instrument and a microwave dataset from the AMSR2 instrument, leading to an improvement in the OSTIA analysis accuracy.
2016/03/23	Problems with the quality of MetOp-B AVHRR data necessitated a reversion to MetOp-A AVHRR in the bias correction reference dataset.

2016/09/05	Reprocessing of analyses for the period 23 February to 7 April 2016 using MetOp-A AVHRR data - instead of MetOp-B AVHRR data - in the bias correction reference dataset. The bias correction reference data are now consistent for the whole period. No MetOp-B AVHRR data are used.
2016/11/08	The satellite reference data used for bias correcting other satellite data was changed to nighttime VIIRS SSTs. This improves the accuracy of the OSTIA analyses.
2018/02/13	The data assimilation scheme used to generate the product was changed to the NEMOVAR variational scheme. The sea ice field included in the files began to be generated using NEMOVAR rather than a regridding of the input observations.
2018/09/25	Water temperature in the Baltic Sea used to infer when ice is present, in order to overcome issues due to sparse ice concentration observations in the region. The method was already being used to infer when there is ice in lakes. It was, however, refined to reduce artefacts that can occur due to interactions between the ice concentration observations and the ice inferred from the water temperature, so both the Baltic ice and lake ice were improved by this change.
2019/03/12	The feature resolution of the analyses were improved. Users will see larger gradients in the SST fields and sharper features. This is illustrated in I.1 below. The set of data sources used to generate the SST analysis was expanded to include SSTs from the Sea and Land Surface Temperature Radiometer (SLSTR) on Sentinel 3A.  **Figure1.1.** Illustration of the change in sharpness in features and increase in the strength of SST gradients due to the 12th March 2019 upgrade.
2019/04/16	The production of monthly and seasonal averages was transitioned into the same operational production system used for the daily SST analyses.
2019/11/27	AVHRR from the NOAA19 satellite is no longer used in the generation of the product but SLSTR from Sentinel 3B and VIIRS from NOAA20 are now used.
2022/11/29	Addition of AVHRR data from the MetOp-C satellite to the collection of data used to generate the product.

2022/12/15	Due to Brexit, since 15th December 2022 indefinitely, only the daily analysis field is provided while all the other datasets are frozen.
2025/06/24	Updated dataset length to be the most recent two years, updated target delivery time and removal of details of datasets that are no longer provided.
2025/05/28	Upstream data change ingestion of NOAA21 VIIRS observations

DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001
Geographical coverage	Global : 180°W-180°E ; 90°S – 90°N
Variables	● Sea surface temperature ● Sea ice concentration
Product Type	Near Real Time - Analysis
Available time series	-2Y to 1 day before real time
Temporal resolution	Daily
Update Frequency	Daily
Target delivery time	12:00 UTC
Horizontal resolution	1/20 degree (regular grid)
Number of vertical levels	Surface only
Format	NetCDF4 CF-1.4

<u>The runtime schedule:</u>
The Operational Sea surface Temperature and Ice Analysis (OSTIA) system is run daily in the operational suite at the UK Met Office. It is run at 05:50 UTC and is usually completed before 06:45 UTC. Products are available to users by 12 :00 UTC at the latest, and normally by 08 :15 UTC.

b) Production System Description

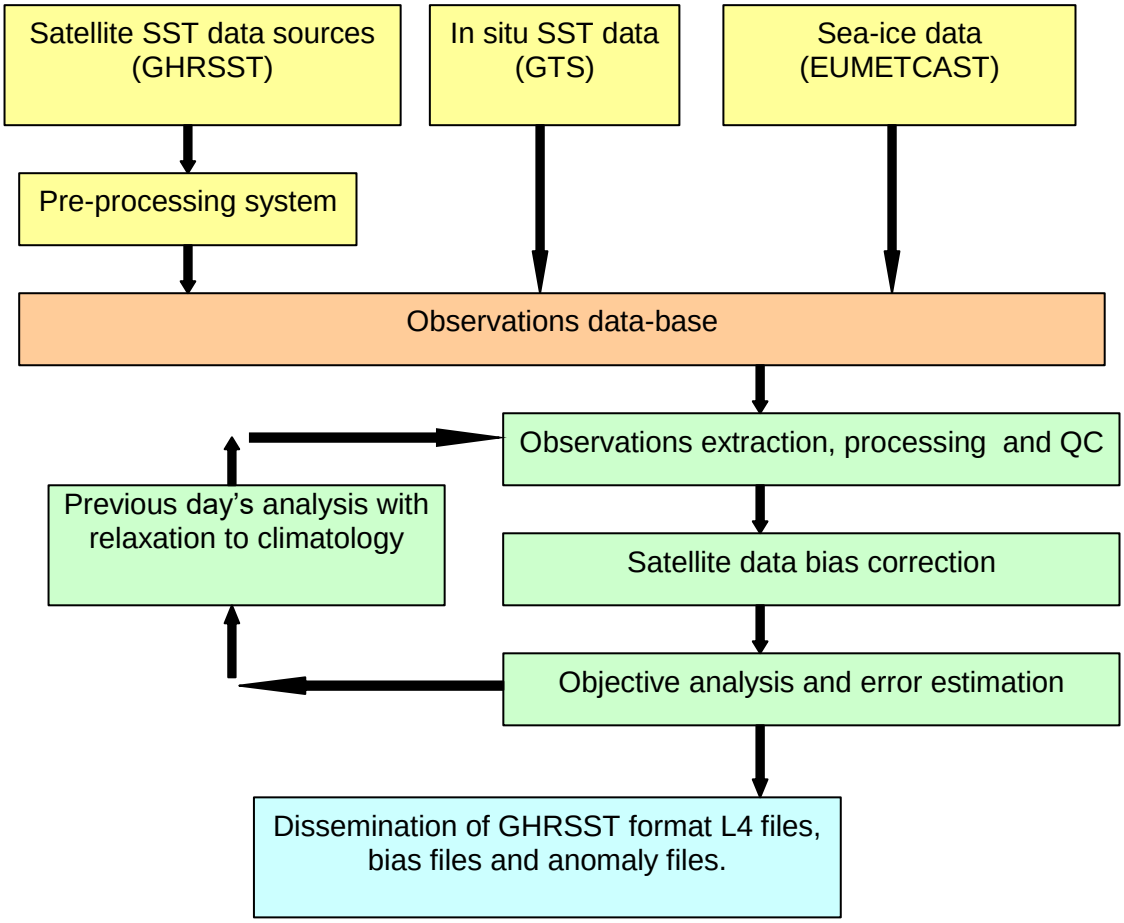


Figure 2.1 Schematic diagram of the operational OSTIA processing chain at the UK Met Office.

Figure 2.1 shows the different steps for the creation of the OSTIA products at the Met Office.

The following inputs are collected for input to OSTIA:

SST satellite data: The original data from different producers (including NASA, NOAA, IFREMER, EUMETSAT OSI-SAF, ESA) in GHRSST compliant format are collected.

All data are provided with Sensor Specific Error Statistics that give an estimate of the systematic and random errors at pixel level. In addition, quality level flags are provided.

The data used in OSTIA include: infra-red data from the AVHRR instruments on board MetOp satellites, SEVIRI on board the MSG satellites, VIIRS data on board NOAA20 and NOAA21 and SLSTR data from Sentinel 3A and Sentinel 3B. Microwave data the AMSR2 instrument on board GCOM-W1 are also used.

In situ SST data from ships, drifting and moored buoys are used in OSTIA.

Ice concentration data from the EUMETSAT OSI-SAF are used in OSTIA for the ocean. Since mid-April 2013 NCEP data have also been used for ice in inland waters.

Climatologies: the SST climatologies used have been derived from the OSTIA reanalysis, produced as part of the MyOcean projects which preceded the Copernicus Marine Service.

Lake Surface Water Temperature (LSWT) data are available within some of the satellite data products. These data are processed using the SST algorithms rather than lake-specific algorithms (which take into account the emissivity and altitude of the lakes) and are therefore less accurate than the data over the open ocean. In situ data over lakes are also available over the GTS, mainly for the Great Lakes.

c) Dataset details

Product Lines	SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001	
Geographical coverage	Global	
Datasets: **METOFFICE-GLO-SST-L4-NRT-OBS-SST-V2** contains: daily foundation SST and sea ice values Variables name in the NetCDF file and Unit: Long_name & Standard_name		
Product Type	Analysis	Forecast
Update frequency	Daily	N/A
Available time series	-2Y - up to real-time	N/A
Target delivery time	Daily at 12pm (noon) UTC	N/A
Temporal resolution	Daily means over a day (midnight to midnight, centered at noon).	

Dataset : SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001 contains daily foundation SST and sea ice values Variables name in the NetCDF file and Unit: Long_name & Standard_name
analysed_sst [K] **analysed sea surface temperature** **sea_surface_foundation_temperature**
analysis_error [K] **estimated error standard deviation of analysed_sst** **sea_surface_foundation_temperature standard_error**
sea_ice_fraction **sea ice area fraction** **sea_ice_area_fraction**
mask **land sea ice lake bit mask**

d) Additional information on some parameters

N/A	N/A

e) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Analysis	N/A	-24 - 0	08am -> 12 noon	Every day

f) Catalogue updating

The product is updated **daily** by **1200 UTC**.

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily	daily means over a day (midnight to midnight, centered at noon).	N/A

h) Product use and impact

This product is aimed at applications requiring gap-free observational sea surface temperature data covering the global ocean and lakes and/or ice concentration.

FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

a. File size

DATASET NAME	NAME OF FILE	DIMENSION [GB]
METOFFICE-GLO-SST-L4-NRT-OBS-SST-V2	YYYYMMDD120000-UKMO-L4_GHRSST-SSTfnd-OSTIA-GLOB-v02.0-fv02.0.nc	0.015

b. Structure of files

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-001.pdf

1) ANNEX

For dataset METOFFICE-GLO-SST-L4-NRT-OBS-SST-V2

```
netcdf \20250101120000-UKMO-L4_GHRSST-SSTfnd-OSTIA-GLOB-v02.0-fv02.0 {
dimensions:
        time = 1 ;
        lat = 3600 ;
        lon = 7200 ;
variables:
        int time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
                time:comment = "" ;
        float lat(lat) ;
                lat:standard_name = "latitude" ;
                lat:long_name = "latitude" ;
                lat:units = "degrees_north" ;
                lat:valid_min = -90.f ;
                lat:valid_max = 90.f ;
                lat:axis = "Y" ;
                lat:comment = " Latitude geographical coordinates,WGS84 projection" ;
        float lon(lon) ;
                lon:standard_name = "longitude" ;
                lon:long_name = "longitude" ;
                lon:units = "degrees_east" ;
                lon:valid_min = -180.f ;
                lon:valid_max = 180.f ;
                lon:axis = "X" ;
                lon:comment = " Longitude geographical coordinates,WGS84 projection" ;
        short analysed_sst(time, lat, lon) ;
                analysed_sst:long_name = "analysed sea surface temperature" ;
                analysed_sst:standard_name = "sea_surface_foundation_temperature" ;
                analysed_sst:units = "kelvin" ;
                analysed_sst:coordinates = "lon lat" ;
                analysed_sst:_FillValue = -32768s ;
                analysed_sst:add_offset = 273.15f ;
                analysed_sst:scale_factor = 0.01f ;
                analysed_sst:valid_min = -300s ;
                analysed_sst:valid_max = 4500s ;
                analysed_sst:source = "AVHRR18_G-NAVO-L2P-V1.0, AVHRR19_G-NAVO-L2P-V1.0,
AVHRR_SST_METOP_B-OSISAF-L2P-V1.0, VIIRS_NPP-OSPO-L2P-V2.3, AMSR2-REMSS-L2P-
V07.2, GOES13-OSISAF-L3C-V1.0, SEVIRI_SST-OSISAF-L3C-V1.0, OSISAF_ICE, NCEP_ICE" ;
                analysed_sst:reference = "Good S, Fiedler E, Mao C, Martin MJ, Maycock A, Reid R,
Roberts-Jones J, Searle T, Waters J, While J, Worsfold M. The Current Configuration of the OSTIA
System for Operational Production of Foundation Sea Surface Temperature and Ice Concentration
Analyses. Remote Sensing. 2020; 12(4):720 https://doi.org/10.3390/rs12040720" ;
                analysed_sst:comment = " OSTIA foundation SST" ;
        short analysis_error(time, lat, lon) ;
                analysis_error:long_name = "estimated error standard deviation of analysed_sst" ;
                analysis_error:standard_name = "sea_surface_foundation_temperature standard_error"
;
                analysis_error:units = "kelvin" ;
                analysis_error:coordinates = "lon lat" ;
```

```
                analysis_error:_FillValue = -32768s ;
                analysis_error:add_offset = 0.f ;
                analysis_error:scale_factor = 0.01f ;
                analysis_error:valid_min = 0s ;
                analysis_error:valid_max = 32767s ;
                analysis_error:comment = " OSTIA foundation SST analysis standard deviation error" ;
        byte sea_ice_fraction(time, lat, lon) ;
                sea_ice_fraction:long_name = "sea ice area fraction" ;
                sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
                sea_ice_fraction:units = "1" ;
                sea_ice_fraction:coordinates = "lon lat" ;
                sea_ice_fraction:_FillValue = -128b ;
                sea_ice_fraction:add_offset = 0.f ;
                sea_ice_fraction:scale_factor = 0.01f ;
                sea_ice_fraction:valid_min = 0b ;
                sea_ice_fraction:valid_max = 100b ;
                sea_ice_fraction:source = "EUMETSAT OSI-SAF" ;
                sea_ice_fraction:comment = " Sea ice area fraction" ;
        byte mask(time, lat, lon) ;
                mask:long_name = "land sea ice lake bit mask" ;
                mask:coordinates = "lon lat" ;
                mask:_FillValue = -128b ;
                mask:valid_min = 1b ;
                mask:valid_max = 31b ;
                mask:flag_masks = 1b, 2b, 4b, 8b, 16b ;
                mask:flag_meanings = "water land optional_lake_surface sea_ice
optional_river_surface" ;
                mask:source = "NAVOCEANO_landmask_v1.0 EUMETSAT_OSI-SAF_icemask
ARCLake_lakemask" ;
                mask:comment = " Land/ open ocean/ sea ice /lake mask" ;

// global attributes:
                :Conventions = "CF-1.4, ACDD-1.3" ;
                :title = "Global SST & Sea Ice Analysis, L4 OSTIA, 0.05 deg daily (METOFFICE-GLO-
SST-L4-NRT-OBS-SST-V2)" ;
                :summary = "A merged, multi-sensor L4 Foundation SST product" ;
                :references = "Good S, Fiedler E, Mao C, Martin MJ, Maycock A, Reid R, Roberts-
Jones J, Searle T, Waters J, While J, Worsfold M. The Current Configuration of the OSTIA System for
Operational Production of Foundation Sea Surface Temperature and Ice Concentration Analyses.
Remote Sensing. 2020; 12(4):720 https://doi.org/10.3390/rs12040720" ;
                :institution = "UKMO" ;
                :history = "Created from sst.nc; obs_anal.nc; seaice.nc" ;
                :comment = "WARNING Some applications are unable to properly handle signed byte
values. If values are encountered > 127, please subtract 256 from this reported value" ;
                :license = "These data are available free of charge under the Copernicus Marine
Service data policy" ;
                :id = "OSTIA-UKMO-L4-GLOB-v2.0" ;
                :naming_authority = "org.ghrsst" ;
                :product_version = "3.5" ;
                :uuid = "536d4865-f5a8-45b2-806f-1f1db491069a" ;
                :gds_version_id = "2.4" ;
                :netcdf_version_id = "4.1" ;
                :date_created = "20250102T063538Z" ;
                :start_time = "20250101T000000Z" ;
                :time_coverage_start = "20250101T000000Z" ;
                :stop_time = "20250102T000000Z" ;
                :time_coverage_end = "20250102T000000Z" ;
                :file_quality_level = 3 ;
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:source = "AVHRR18_G-NAVO-L2P-V1.0, AVHRR19_G-NAVO-L2P-V1.0, AVHRR_SST_METOP_B-OSISAF-L2P-V1.0, VIIRS_NPP-OSPO-L2P-V2.3, AMSR2-REMSS-L2P-V07.2, GOES13-OSISAF-L3C-V1.0, SEVIRI_SST-OSISAF-L3C-V1.0, OSISAF_ICE, NCEP_ICE" ;
:platform = "NOAA-18, NOAA-19, MetOpB, NPP, GCOM-W, GOES13, MSG4, DMSP-F17, DMSP-F15" ;
:sensor = "AVHRR, VIIRS, AMSR2, GOES_IMAGER, SEVIRI, SSMIS, SSM/I" ;
:Metadata_Conventions = "Unidata Observation Dataset v1.0" ;
:metadata_link = "http://podaac.jpl.nasa.gov/ws/metadata/dataset?format=gcmd&shortName=UKMO-L4HRfnd-GLOB-OSTIA" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention" ;
:westernmost_longitude = -180.f ;
:easternmost_longitude = 180.f ;
:southernmost_latitude = -90.f ;
:northernmost_latitude = 90.f ;
:spatial_resolution = "0.05 degree" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lat_resolution = 0.05f ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_lon_resolution = 0.05f ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: These data were provided by GHRSST, Met Office and the Copernicus Marine Service" ;
:creator_name = "Met Office as part of the Copernicus Marine Service" ;
:creator_email = " https://marine.copernicus.eu/contact" ;
:creator_url = " https://marine.copernicus.eu" ;
:project = "Group for High Resolution Sea Surface Temperature" ;
:publisher_name = "GHRSST Project Office" ;
:publisher_url = "http://www.ghrsst.org" ;
:publisher_email = "ghrsst-po@nceo.ac.uk" ;
:processing_level = "L4" ;
:cdm_data_type = "grid" ;
}